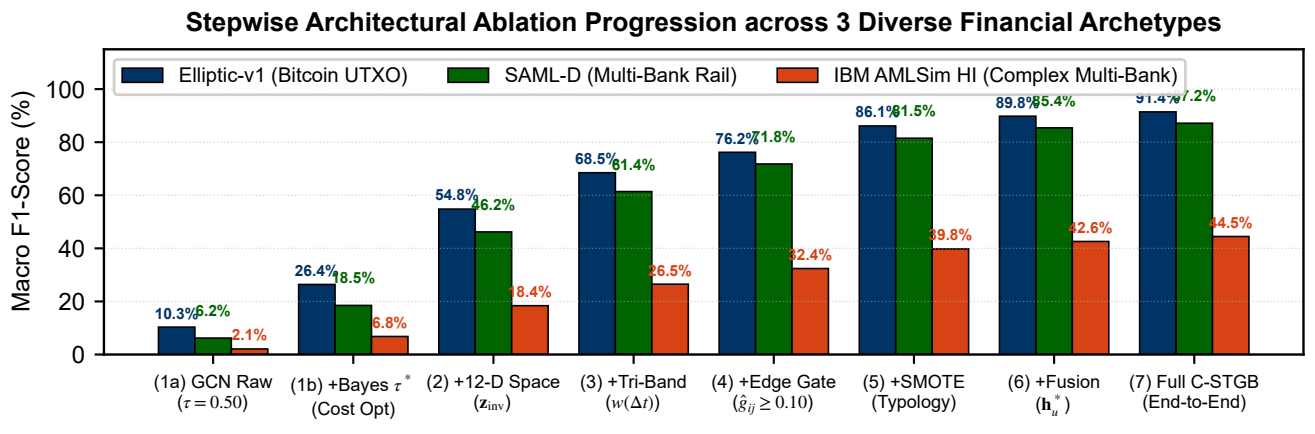

Stepwise Architectural Ablation Progression across 3 Diverse Financial Archetypes
Elliptic-v1 (Bitcoin UTXO)
SAML-D (Multi-Bank Rail)
IBM AMLSim HI (Complex Multi-Bank)
Macro F1-Score (%)
0
20
40
60
80
100
10.3%
6.2%
2.1%
26.4%
18.5%
6.8%
54.8%
46.2%
18.4%
68.5%
61.4%
26.5%
76.2%
71.8%
32.4%
86.1%
81.5%
39.8%
89.8%
85.4%
42.6%
91.4%
87.2%
44.5%
(1a) GCN Raw
($\tau = 0.50$)
(1b) +Bayes τ^*
(Cost Opt)
(2) +12-D Space
($\mathbf{z}_{\text{inv}}$)
(3) +Tri-Band
($w(\Delta t)$)
(4) +Edge Gate
($\hat{g}_{ij} \geq 0.10$)
(5) +SMOTE
(Typology)
(6) +Fusion
($\mathbf{h}_u^*$)
(7) Full C-STGB
(End-to-End)